

Table 2

Code	Priority	Severity
0	<i>Emerg</i>	System is unusable
1	<i>Alert</i>	Action must be taken immediately
2	<i>Crit</i>	Critical condition
3	<i>Err</i>	Non-critical error condition
4	<i>Warning</i>	Warning condition
5	<i>Notice</i>	Normal but significant event
6	<i>Info</i>	Informational event
7	<i>Debug</i>	Debugging level message

The *rsyslogd* service uses the facility and prioritisation of log messages to determine how to handle them. This is configured by the file */etc/rsyslog.conf* and by **.conf* files in */etc/rsyslog.d*. Programs and administrators can change *rsyslogd* configuration in a way that it will not be overwritten by updates to *rsyslogd* by putting customised files with a *.conf* suffix in the */etc/rsyslog.d* directory.

The *Rules* section of */etc/rsyslog.conf* contains directives that define where log messages are saved. The left side of each line indicates that the facility and severity of the log message directive matches. The *rsyslog.conf* file can utilise the character *** as a wild card in the facility and the severity fields, where it either stands for all the facilities or all the severities. The right side of each line indicates what file the log message is to be saved in. Log messages are always saved in files in the */var/log* directory.

Note: Log files are maintained by the *rsyslog* service, and the */var/log* directory contains a variety of log files specific to certain services. For example, the Apache Web server or Samba write their own log files into a corresponding sub-directory of the */var/log* directory.

A message handled by *rsyslog* can appear in multiple different log files. To prevent this, the severity field can be set to *None*, which means that none of the messages directed to this facility are added to a specific log file.

Instead of logging syslog messages to a file, they can be printed to the terminals of all logged-in users. In the default *rsyslog.conf* file, this is done for all messages that have *emerg* priority. A sample of the *Rules* section of *rsyslog.conf* is shown in Figure 1.

Note: The *rsyslog.conf* file is documented by the man page of *rsyslog.conf* (5) and there is extensive documentation in */usr/share/doc/rsyslog-*/manual.html* in Red Hat Enterprise Linux 7.

Log file rotation

Logs are rotated by the *logrotate* utility to keep them from

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#### RULES ####

# Log all kernel messages to the console.
# Logging much else clutters up the screen.
#kern.*                                          /dev/console

# Log anything (except mail) of level info or higher.
# Don't log private authentication messages!
*.info;mail.none;authpriv.none;cron.none      /var/log/messages

# The authpriv file has restricted access.
authpriv.*                                     /var/log/secure

# Log all the mail messages in one place.
mail.*                                          -/var/log/maillog

# Log cron stuff
cron.*                                         /var/log/cron

# Everybody gets emergency messages
*.emerg                                         :omusrmsg:*

# Save news errors of level crit and higher in a special file.
uucp,news.crit                                 /var/log/spooler

# Save boot messages also to boot.log
local7.*                                       /var/log/boot.log

```

Figure 1: Default *Rules* section of the *rsyslog.conf* file

filling the file system containing */var/log*. When a log file is rotated, it is renamed with an extension indicating the date on which it was rotated — the old */var/log/messages* file may become */var/log/messages-20180630* if it is rotated on June 30, 2018. Once the old log file is rotated, a new log file is created and the service that writes to it is notified.

After a certain number of rotations, typically after four weeks, the old log file is discarded to free disk space. A cron job runs the *logrotate* program daily to see if any logs need to be rotated. Most log files are rotated weekly, but *logrotate* rotates some faster or slower, or when they reach a certain size. Configuration of *logrotate* is not covered in this article but you can check out the *logrotate* (8) man page.

Analyse a syslog entry

The system log written by *rsyslog* starts with the oldest message on the top and the newest message at the end of the log file. All log entries in the log file managed by *rsyslog* are recorded in a standard format. An example is shown below, which is a sample taken from the */var/log/secure* log file.

- Feb 21 20:11:48: Indicates the time stamp when the log entry was recorded.
- localhost: Indicates the host from where the log message was sent.
- sshd[1433]: Indicates the program or process that sent the log message.
- Failed password for student from 172.25.8.10 port 59344 ssh2: Indicates the actual message sent.

Monitoring a log file with a tail

To reproduce problems and issues, it is helpful to monitor more log files for events. The *tail -f /path/to/file* command outputs the last ten lines of the file specified, and continues to output new lines as they get written to the monitored file.

To monitor for failed login attempts on one terminal, run *ssh* from one machine to another with the wrong password and list it as shown in Figure 2.